



CostVision

The AI cost-intelligence platform

From a CAD model, a board photo or a plain description — to a defensible, auditable, negotiation-ready should-cost, in minutes. And it gets smarter with every real quote.

● CAD → Cost

● PCB Image → BOM

● 19 Commodity Engines

● Agentic AI

Executive briefing & live demo · July 2026 · Key figures verified from live runs; illustrative values marked

Executive summary — the whole platform on one page



What it is: one AI platform that turns any engineering input — a CAD model, a board photo, a description — into a defensible, auditable should-cost in minutes, prices it across 20 regions, hands us the negotiation, and learns from every real quote. AI understands the input; deterministic maths sets every price.



Bottom line: faster quotes, better margins, one consistent number — and a proprietary cost-intelligence asset that compounds, on our own infrastructure.

Background & agenda

The picture: we already run rigorous should-costing across every commodity with our internal tool — a proven, bottom-up method. Today it is hands-on: every input is selected and keyed in by hand from the 3D CAD and drawings. CostVision is the next step — it keeps that trusted method and automates the manual work with an AI layer: reading the CAD, suggesting material & process, and keeping rates current. Same defensible number, in minutes — and the knowledge compounds as a company asset.

- 1 From manual should-costing to AI-assisted
- 3 CAD → Cost — workflow, AI & accuracy
- 5 Global should-cost — engines, rates & transparency
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- 9 Way forward, blueprint & live demo

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- 8 Trust, accuracy & data security

From manual should-costing to AI-assisted

We already do rigorous should-costing across every commodity. CostVision keeps that method — and automates the hands-on steps with AI.

OUR SHOULD-COST TODAY · capable, but hands-on

- **Trusted & thorough**
a rigorous bottom-up method across every commodity
- **Hands-on setup**
material, process & machine selected manually, part by part
- **Manual data entry**
geometry & tolerances read off 3D CAD and drawings by hand
- **Updated by hand**
rates & inputs refreshed manually as economics change



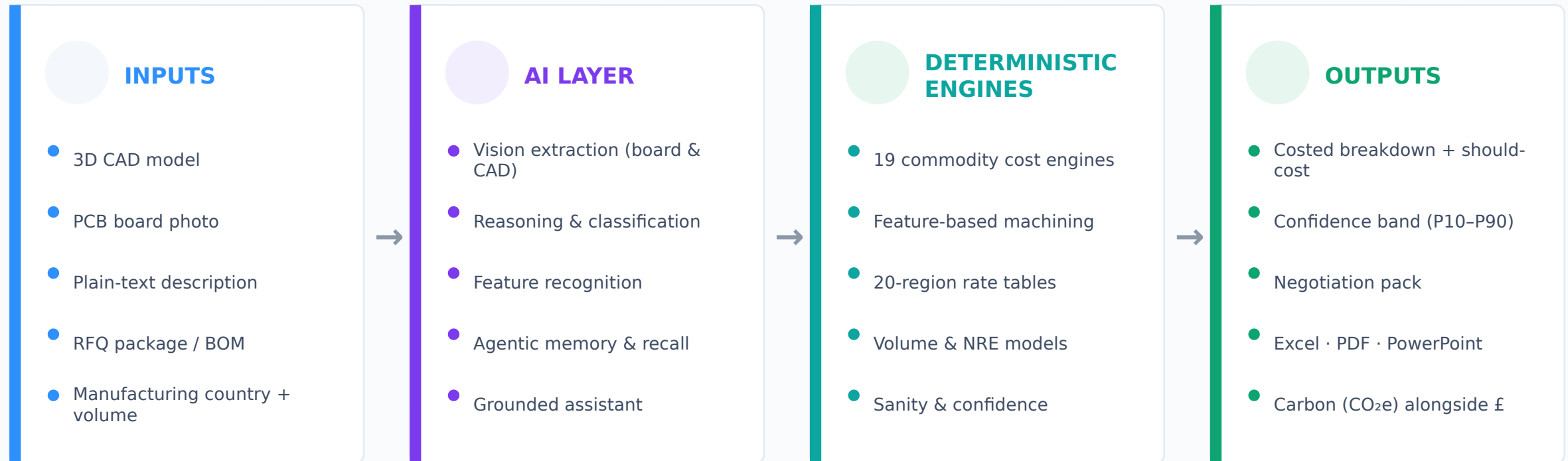
WITH COSTVISION · automated & AI-assisted

- ✓ **Same rigour, automated**
the deterministic method we trust — the manual steps removed
- ✓ **Reads CAD & drawings**
geometry, features & tolerances extracted automatically
- ✓ **AI suggests, we decide**
material & process proposed with a confidence score to confirm
- ✓ **One versioned rate library**
upload updated rates once — every part re-prices in a click

The step forward: the same defensible should-cost, with the manual work automated and AI-assisted — so our experts spend their time on decisions, not data entry.

CostVision at a glance — the platform

Any engineering input → AI understands it → deterministic engines price it → we get a defensible, negotiation-ready result.



The golden rule: AI only ever understands the input. It never sets the price — every pound is deterministic, auditable arithmetic we can trace and defend.

One platform, four jobs — the capability map

ESTIMATE

- CAD → Cost (3D model)
- 19 commodity cost engines
- Detailed teardown
- PCB Image → BOM
- Feature-based machining

UNDERSTAND

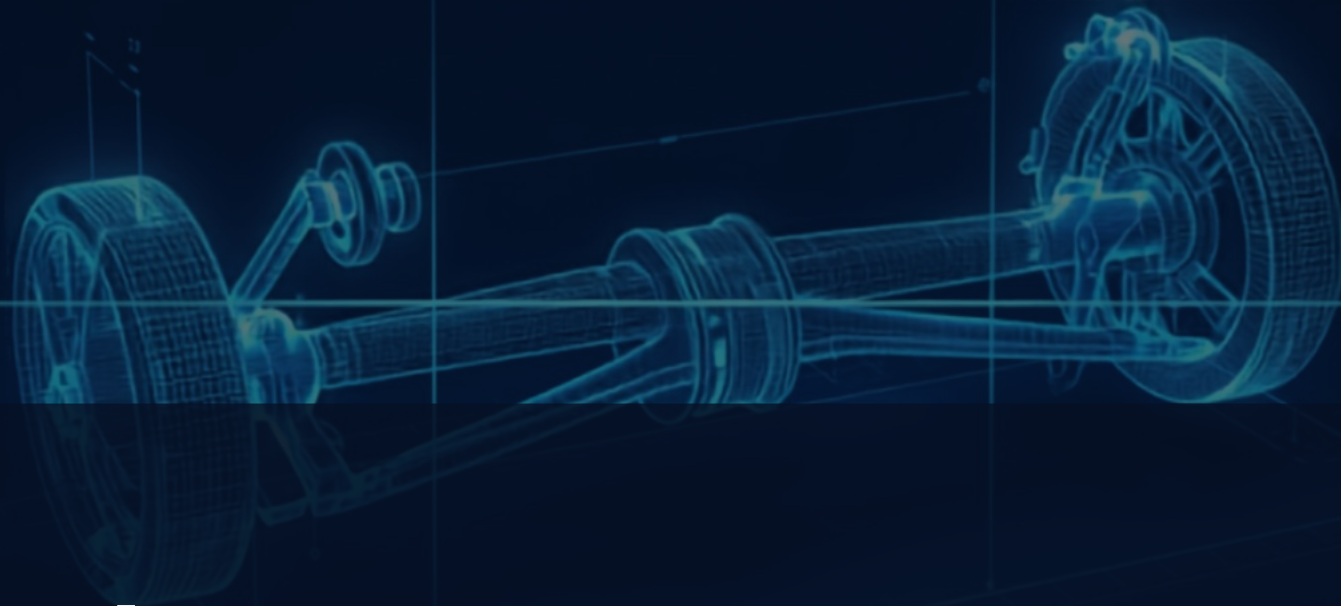
- 3D CAD viewer + B-rep AI
- Grounded AI assistant
- Carbon co-costing
- Geometry feature table
- RFQ analyst

DECIDE & NEGOTIATE

- Negotiation Intelligence
- Volume & what-if scenarios
- Causal negotiation coach
- 20-region should-cost
- Conformal confidence bands

LEARN & GOVERN

- Agentic memory & recall
- Autonomous savings agent
- On-premise & secure
- Self-correction from quotes
- Full audit trail / provenance



DEEP DIVE

CAD → Cost

CAD → Cost — from a 3D model to a should-cost

Drop in a 3D model and CostVision reads the geometry, recognises the manufacturing features, and prices the part — with every number traced back to the model.



Reads the real geometry

A true geometry engine (OCCT / STL) measures volume, bounding box and surface area — measured ground truth, not a guess.



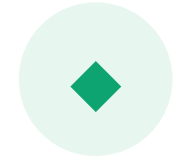
Recognises features

Holes, pockets, slots, bends and faces are detected and turned into machining operations automatically.



Suggests material & process

AI proposes the material family and process route with a confidence score — we stay in control.



Prices it — and explains

A deterministic 8-bucket engine (material → margin) returns a full should-cost with a confidence band and DFM guidance.

Add a 2D drawing to sharpen it: attach the engineering-drawing PDF and the AI reads the tolerances & GD&T — tight tolerances pull in grinding/reaming and better surface finish, so the process route and the cost get more accurate. Formats: STEP · IGES · STL, plus an optional 2D drawing.

CAD → Cost — the workflow, end to end



✦ **AI understands the model**

⚙️ **Deterministic maths sets the price**

The safety net is the sanity check. Because the geometry engine gives us measured volume, weight and area, every AI-suggested number is cross-checked against physical reality before it reaches the cost. A hallucinated dimension or an impossible weight is flagged — not silently priced. That single step is why the output is defensible.

CAD → Cost — the AI capabilities



Interactive 3D CAD viewer

Rotate, section and measure the part in the browser — with on-model highlights that link geometry to cost.



B-rep face intelligence

Understands the boundary representation — faces, edges and their relationships — not just a triangle soup.



Geometry feature table

Every detected feature (hole depths, pockets, slots) listed and auto-fed into the costing as real operations.



Feature-based machining

Secondary operations are costed per feature and per commodity — facing, drilling, pockets, bends.



AI material & process advice

Suggested material family and process route, each with an explicit confidence score we can accept or override.



DFM & manufacturability

A manufacturability score plus design-for-manufacture flags that point at what is driving the cost.

Model tiers: Haiku classifies the commodity · Sonnet 5 runs the specialist analysis · Opus 4.8 on the "deep analysis" toggle for the hardest parts.

CAD → Cost — how we make it accurate

Accuracy is engineered in layers — measured truth, deterministic maths, and a check on every AI claim before it can touch the price.

1 Measured ground truth — A true geometry engine (OCCT) measures volume, weight and area — the AI is checked against physics, not trusted blindly.

2 Bounded sanity checks — Volume drift over 25% is flagged and over 50% rejected; weight is cross-checked against measured density — hallucinations never reach the cost.

3 Deterministic 8-bucket engine — No AI in the money maths — it reproduces a hand-calculated £23.27 benchmark part to within 0.01%, and returns the same cost every run.

4 Provenance on every number — Each figure traces to its source — geometry, rate table or AI suggestion — with a High/Medium/Low confidence, defensible line by line.

5 Confidence, not false precision — Every estimate carries a low-mid-high band and a manufacturability score, never a fake single number.

6 Proven on real parts — Demonstrated end-to-end on real automotive parts — from a Tata Altroz bumper to premium-SUV components — through the live system.

Why it earns trust

The AI can suggest — it can never quietly decide the number.

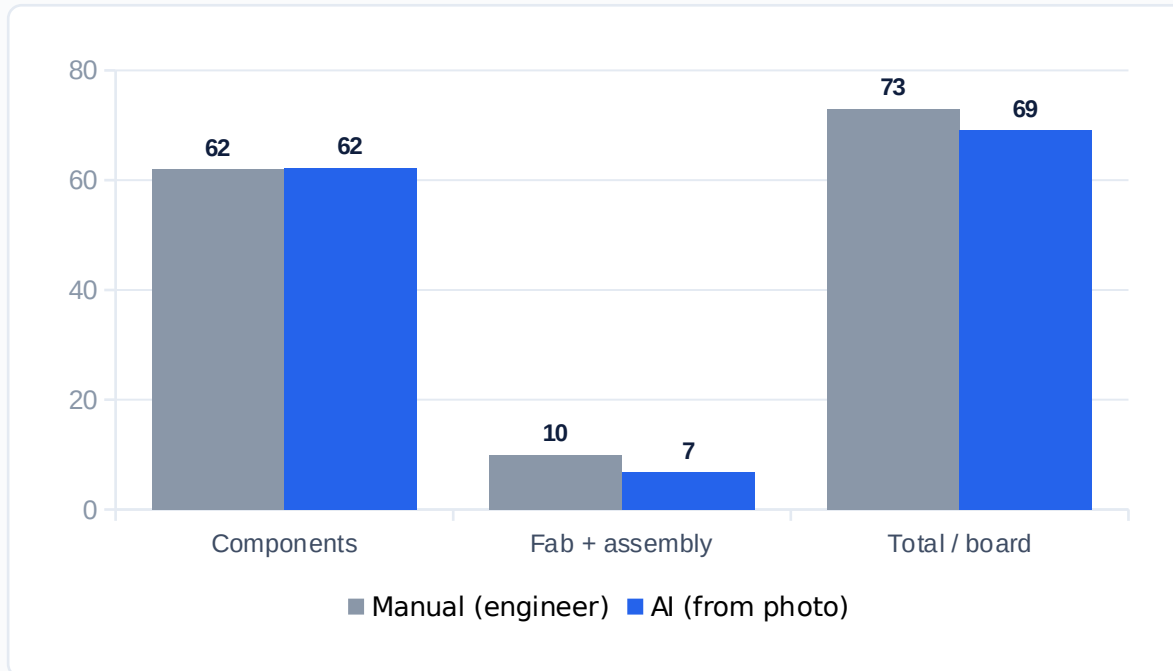
Because the geometry is measured and the maths is deterministic, a wrong guess gets caught and flagged, not baked into a quote. That is the difference between a demo and a number we can put in front of a supplier.

ONE MORE INPUT · ELECTRONICS

PCB photo → BOM

PCB photo → BOM → should-cost

The same idea for electronics: photograph a circuit board and get a costed, flagged bill of materials in ~60 seconds.



What it delivers

- ✓ **Full BOM from a photo** — every component identified, OCR of IC markings
- ✓ **Confirmed vs to-verify** — headline splits the £ we can trust from the £ to firm up
- ✓ **Guardrails on misreads** — a mis-read part is capped & flagged, never left to dominate
- ✓ **Regional should-cost** — fab + assembly + BOM priced across regions
- ✓ **ASIL & compliance aware** — ISO 26262 safety class inferred automatically

Real automotive ECU, China @ 10k/yr — AI within ~5% of a half-day manual estimate, in ~60 s.

Verified live: that ECU came back ASIL-B, 23 BOM lines, £69.11/board — within ~5% of a manual estimate, in ~60 seconds. A full deep-dive deck exists for this feature.

Estimate anything — 19 commodity engines

Purpose-built, bottom-up models for the whole part portfolio — not one generic formula stretched over everything.

● Casting — sand / gravity / investment

● HPDC die-casting

● CNC machining

● Forging

● Sheet metal — stamping

● Sheet-metal fabrication

● Injection moulding

● Blow moulding

● Thermoforming

● Rotational moulding

● Rubber & elastomers

● Composites

● Extrusion

● Wiring harness

● PCB fabrication

● PCBA — electronics assembly

● Painting & surface finish

● Body-in-white assembly

One method under all of them: the same auditable 8-bucket cost stack (material → margin) with feature-based machining on top. The same part then prices across 20 regions — that's the next section.

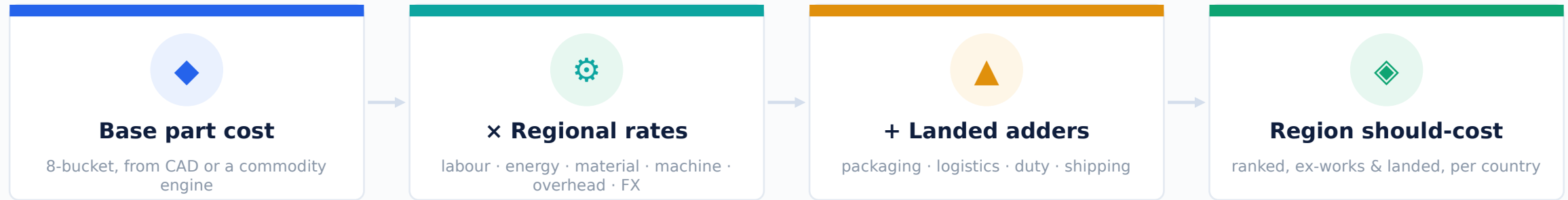


DEEP DIVE · SHOULD-COST MODEL

Global should-cost

Global should-cost — how the country model works

One base cost, re-priced with each region's real economics — deterministic, transparent and landed-cost aware. No AI in the money.



What we give it

- ✓ **The part** — once — from a CAD model, a photo, or a commodity form
- ✓ **The regions** — pick any of the 20, or compare them all
- ✓ **Volume & Incoterm** — drives volume curve and landed vs ex-works

What is factored in, per region

- Labour by trade (£/hr)
- Material factors (resin & metal)
- Overhead multiplier
- FX to GBP
- Energy — elec & gas (£/kWh)
- Machine-rate multiplier
- Packaging & logistics
- Duty & shipping adders

The actual rates — full transparency, no black box

These are the real 2026-Q2 rates every costing uses — labour, energy and the regional multipliers. Nothing is hidden or hand-waved.

Region	Skilled £/hr	Foundry £/hr	Elec £/kWh	Machine ×	Overhead ×	Logistics ×
United Kingdom	26.00	18.50	0.23	1.00	1.00	1.00
Germany	40.50	28.00	0.20	1.05	1.10	1.15
United States	34.00	24.00	0.10	0.85	0.95	1.25
Poland	12.00	8.00	0.14	0.72	0.85	1.20
Czech Rep.	13.00	9.50	0.13	0.74	0.87	1.20
Mexico	7.50	4.80	0.08	0.60	0.78	1.35
Turkey	6.50	4.80	0.09	0.60	0.78	1.30
China	7.90	5.00	0.07	0.55	0.75	1.45
India	5.10	3.00	0.07	0.52	0.72	1.50
Vietnam	3.80	2.50	0.06	0.52	0.70	1.50

Showing 10 of 20 regions. Sourced from IPC / industry rate data. Material factors, FX, energy-gas, packaging and duty adders sit behind these too — one versioned library, used by every single costing.

Bring our own rate database — our rates, our numbers

Upload our organisation's own cost rates as one Excel workbook. Every costing then runs on our numbers — validated on upload, versioned, and fully auditable. No black box, no lock-in to our defaults.

Materials

£/kg, scrap-recovery £/kg, density, region — with effective date

Machines

Full rate build-up (deprec., maint., energy, floor, finance, hours, utilisation) → £/hr

Labour

Fully-loaded £/hr by region and skill level

Energy

Electricity & gas £/kWh by region

FX

Currency pairs and exchange rates

Overhead

Overhead % and margin % by commodity and supplier tier

Six rate tables — one tab each in the template.

WHO CAN UPLOAD

Admin-gated. Admins upload, switch company ↔ built-in, or reset. Everyone else costs on the active set — a badge shows **“Company rates active”** vs built-in. One consistent number across the team.

THE EXCEL TEMPLATE

One workbook, 6 tabs. Download → fill → upload. The server validates every row and rejects bad cells with line-level errors. Each row carries an **effective date + confidence**.

UPDATE IN ONE CLICK

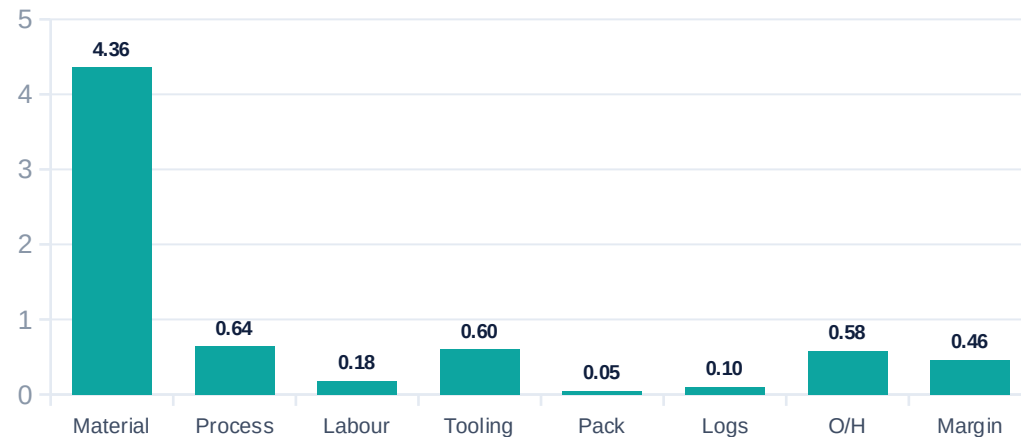
Load a new dated rate set and activate — the whole portfolio re-prices in a click. The library is **versioned**, so we keep current vs old vintages and scenarios flag if costed on an older version.

Our rates, our defensible numbers. Quick to load (download → fill → upload), validated on the way in, versioned and auditable — and every costing across the platform immediately uses them.

Live example — Aluminium HPDC casting, priced everywhere

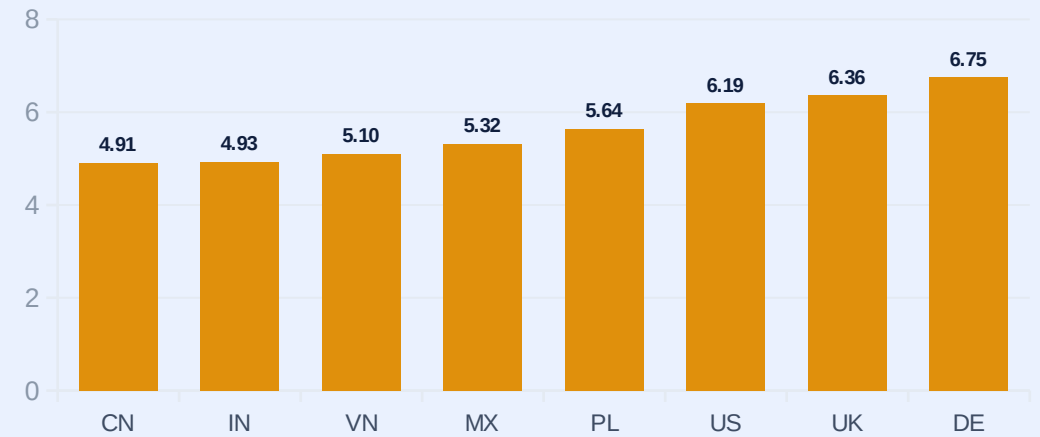
A simple 1.2 kg ADC12 aluminium die-casting (2-cavity, 800T), run through the real engine — UK breakdown and the regional spread.

UK cost stack — £/part



Total £6.96/part · material is 63% — the true driver.

Ex-works £/part by region



China —23% ex-works, narrowing to —12% landed once logistics & duty are added — the model shows both.

🔄 Always current — one versioned rate library

Material factors, labour, energy and overhead all live in one versioned rate library — not scattered across spreadsheets. **Upload an updated, dated rate set and every part re-prices — one click, no re-keying, no stale spreadsheets.** Keeping should-costs current with the latest economics is a single upload, not a manual rebuild.

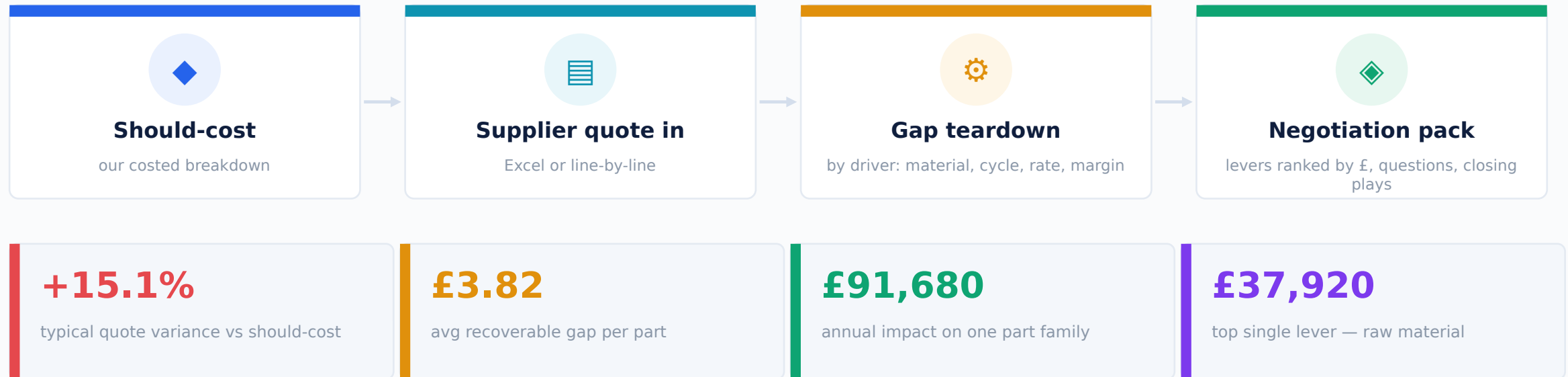


DECIDE & NEGOTIATE

Negotiation Intelligence

Negotiation Intelligence — turn should-cost into savings

A should-cost is useful. A should-cost that walks into a negotiation and wins money back is the product.



Illustrative figures — the same teardown runs on our real supplier quotes.

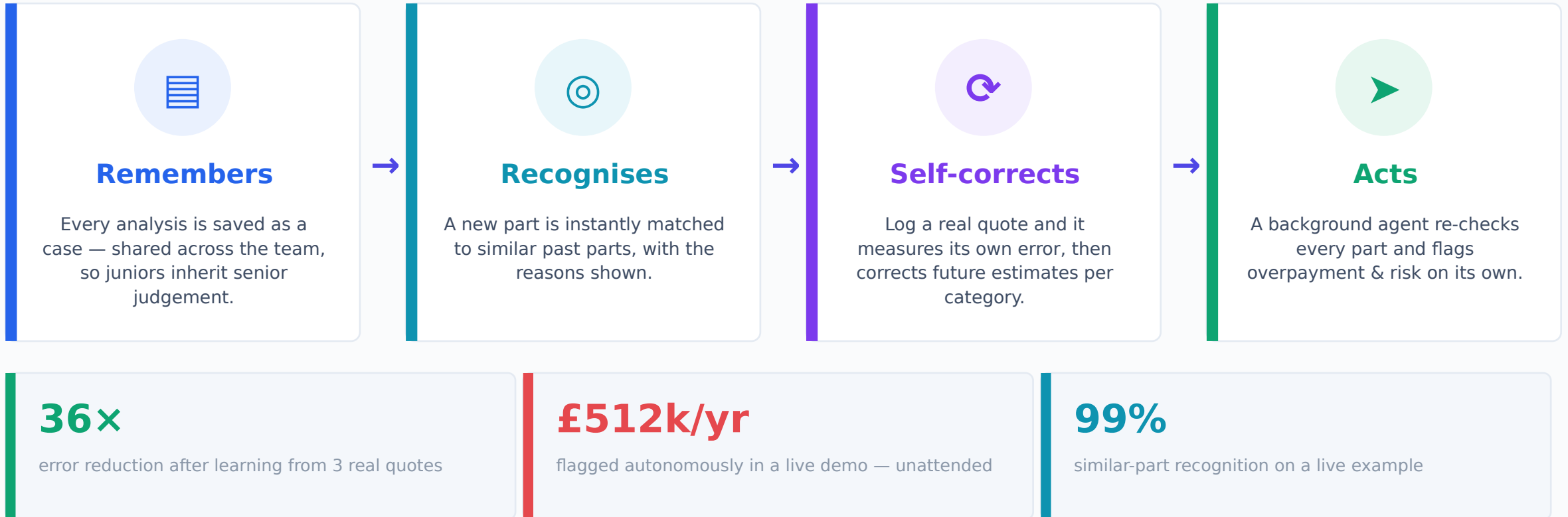
The pack contains: a line-by-line gap table, cost-driver decomposition (which parameter drives each gap), recovery levers ranked by £/yr, ready-to-send supplier questions, and a one-click PDF & PowerPoint. We stop arguing about the total and start negotiating the drivers.



LEARN & GOVERN

Agentic AI

Agentic AI — it learns, remembers and acts



Advanced intelligence — beyond the estimate



Conformal confidence bands

A statistically-guaranteed range proven against our own logged quotes — "90% land within $\pm 6.5\%$ ", evidence not assertion.



Causal negotiation coach

Knows why a part costs what it does and writes the argument: "every 1% move in aluminium shifts price by £0.11".



What-if scenario engine

Move a commodity $\pm 20\%$ and see the price recompute — for one part or the whole portfolio, instantly.



Carbon co-costing

Every part gets a CO₂e figure alongside the £ — increasingly required in automotive & aerospace RFQs.

Each of these turns a number into a decision — a defensible range, a negotiating sentence, a risk pre-empted, a carbon figure.

Trust by design — glass-box, measured, tested



Glass-box, never black-box

Every learned or derived number stays auditable — a value an engineer can read and defend. AI narrates; it never sets the price in secret.



Measured, not asserted

Accuracy is checked against real geometry and real quotes, and reported openly — before and after.



Honest about uncertainty

Confidence bands and clear flags on anything unverified — never a false single number.



Production-grade

1,005 automated tests across 86 suites, exercised end-to-end — and the container build is verified in CI.

The through-line of the whole platform: the AI does the understanding; deterministic maths does the money; and every number is traceable, bounded and testable. That is what lets us put a CostVision number in front of a supplier — or a board.

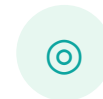
Built to stay right — quality that's enforced, not promised

It isn't enough to be right once. These guardrails run on every change, so the tool stays defensible as it evolves — quality is enforced by the build, not by memory.



The golden rule, machine-enforced

"AI never sets a price" is now a build-time check: if any AI dependency ever reaches the deterministic cost engine, the build fails. It cannot be bypassed.



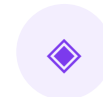
Accuracy measured every build

CAD feature detection is scored against hand-verified golden parts — precision, recall, F1 (≈ 0.96 today). A floor gate catches any drop before it ships.



Every change guarded

1,005 automated tests, a self-audit re-checking every estimate, plus visual-regression screenshots — a regression fails the build, not the demo.



Confidence on every number

Each should-cost shows a P10-P50-P90 band, colour-graded by width — an honest range a buyer can defend, never a false single figure.

Bottom line: the number we put in front of a supplier is defensible by construction — and stays that way. Quality isn't reviewed in after the fact; it's enforced on every commit.

The engineering behind it — a real, in-house asset

Not a prototype or a thin wrapper around a chatbot — a substantial, tested, full-stack platform, custom-built in-house (AI-accelerated).

~75,000

lines of custom code, built in-house

1,005

automated tests · 86 suites

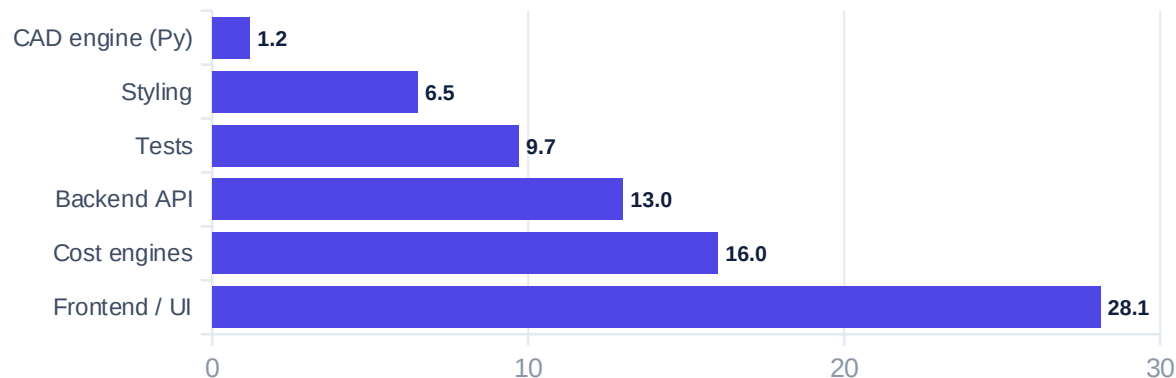
19

purpose-built cost engines

5

languages & runtimes in the stack

Where the code lives — thousands of lines



What that represents

- ◆ **A 16,000-line deterministic cost engine** — the auditable "money maths" — 19 commodities, 20 regions
- ◆ **A real Python geometry kernel** — OCCT/CadQuery reads STEP, IGES & STL — genuine CAD engineering
- ◆ **Full-stack, end to end** — TypeScript backend & UI, Three.js 3D, SQLite, Python — one system
- ◆ **Quality built in** — ~10,400 lines of tests, 1,005 passing — it does not silently regress

The point: this depth of engineering is exactly why the intelligence is a durable, proprietary company asset — not a subscription that can be switched off.

What this means for the business



Speed

Weeks of RFQ cycle collapse to minutes — the first defensible number arrives while the meeting is still happening.



Margin recovery

Line-level should-cost lets us challenge every quote by driver — the agent alone surfaced ~£512k/yr in a demo.



Consistency & speed-to-competence

One consistent answer for everyone; juniors cost like seniors from day one.



A compounding asset

Costing knowledge becomes proprietary data on our own servers — it grows, and competitors cannot buy it.

Net: faster quotes, better margins, consistent numbers, and an intelligence asset that appreciates with every use — at zero extra licence cost, on our own infrastructure.

Way forward — the implementation blueprint

A low-risk, staged rollout — prove it on a pilot, scale to the team, then embed it in how we quote and negotiate.

Phase 1 — Pilot

Weeks 0-6

- ✓ Stand up on-premise
- ✓ Seed memory with historic quotes
- ✓ Cost 20-30 live parts
- ✓ Baseline accuracy vs actuals



Phase 2 — Scale

Weeks 6-16

- ✓ Roll out to costing & sourcing
- ✓ "Log Actual £" as routine
- ✓ Wire into RFQ workflow
- ✓ Monthly agent findings review



Phase 3 — Embed

Quarter 2+

- ✓ Negotiation packs as standard
- ✓ Portfolio what-if & risk
- ✓ Live distributor / market feeds
- ✓ KPI: £ saved & cycle-time

The one dependency, stated honestly: the intelligence starts empty and grows with use. Seeding it with our historic quotes in Phase 1 means it starts smart, not from zero.

Data security & deployment

Built for enterprise from the ground up — our data, our infrastructure, our IP.



On-premise deployment

Runs inside our own walls — the knowledge base and every analysis are stored on our servers and never retained by any vendor.



Keys server-side only

The AI key lives server-side (.env) — the browser never sees it. AI calls proxy through our server to the model endpoint.



Private & air-gapped ready

Private routing (ANTHROPIC_BASE_URL) or air-gapped mode keeps even the AI request in-network for the most sensitive programmes.



Full audit trail

Every number carries provenance; every analysis and correction is logged — defensible and traceable.



Our IP, not a vendor's

The accumulated costing intelligence is a proprietary company asset — it does not walk out with a subscription.



Access & governance

Role-based access, on our identity and infrastructure, under our data-governance policy.

What actually leaves the building: only the AI inference request, sent via our own server and not retained by the model provider — and in air-gapped / private-routing mode, not even that. Our data, our infrastructure, our IP.

Live demo — the 5-minute flow

A tight, high-impact path through the live system — click order, what to show, and the moment that lands.

1	CAD → Cost	Upload a STEP model → show the 3D viewer, geometry & feature table → Calculate	→ the should-cost + confidence band appear in ~30-60 s
2	Add a 2D drawing	Attach the engineering-drawing PDF → re-analyse	→ tolerances tighten the process route — accuracy improves live
3	Compare countries	Switch the region selector / open the comparison	→ the 20-region spread — China vs Germany — in one click
4	PCB photo → BOM	Drop a board photo → run analysis	→ a full costed BOM builds from a single picture
5	Negotiation	Paste a supplier quote → generate pack	→ the gap teardown by driver + ready-to-send questions
6	Autonomous agent	Open the findings panel	→ ~£512k/yr surfaced by itself — the "wow" close

Before we start

- ✓ A sample STEP part loaded
- ✓ A 2D drawing PDF to hand
- ✓ A board photo ready
- ✓ A supplier quote to paste
- ✓ Seeded memory for the agent
- ✓ Cached runs as a fallback

Short on time? Do steps 1, 3 and 6.

Any input in. A defensible, self-improving should-cost out.

Fast · explainable · self-improving · secure — and entirely ours.

THE ASK — THREE SMALL DECISIONS

1

Approve a 6-week on-premise pilot

seed it with our historic quotes

2

Pick a first part family

a CAD part, a board, or a live RFQ

3

Put agent findings on the agenda

it is already finding money

Thank you · happy to take questions, or run the live system now